PAPER_351 ASASSN-14li Tidal Disruption Event: Ultrafast Outflow F_U_Bi_i and Kozima LENR Force

Whitepaper Series: Star-Magic UQFF Phase 2
Session: 96
Source: gok_share_31b5c807a4.txt (Supplemental Gap Analysis Block)
Classification: FIRST UQFF F_U_Bi_i for a TDE with 0.3c ultrafast outflow and Kozima LENR coupling
Author: Daniel T. Murphy

Abstract

ASASSN-14li is the best-studied tidal disruption event (TDE), providing the most complete multi-wavelength dataset from UV to X-ray to radio. The UQFF buoyancy-unified force is computed for the stellar mass black hole remnant (M_BH = 106 M?), yielding F_U_Bi_i -8.32×10 N six orders of magnitude smaller than AGN-scale F_U_Bi_i, reflecting the much smaller BH mass. The ultrafast outflow at v_out = 0.3c is connected to UQFF via the Kozima LENR force component F_Kozima = 10 N at the stellar disruption interface.

---

2. Core Physics

2.1 UQFF Buoyancy-Unified Force (TDE Scale)

$$F_{U\_Bi\_i} \approx -8.32 \times 10^{211}\ \mathrm{N}$$

The six-order-of-magnitude reduction from the AGN scale (-8.32×107 N) reflects M_BH = 106 M? vs. 10? M?.

2.2 Ultrafast Outflow

$$v_{\text{out}} = 0.3c = 9.0 \times 10^{7}\ \mathrm{m/s}$$

Observed in Chandra/XMM-Newton blueshifted Fe K absorption lines. The UQFF kinetic coupling:

$$P_{\text{outflow}} = \frac{1}{2}\dot{M}_{\text{out}} v_{\text{out}}^2 = \frac{1}{2}\dot{M}_{\text{out}}(0.3c)^2$$

2.3 Kozima LENR Force Component

$$F_{\text{Kozima}} = 1 \times 10^{30}\ \mathrm{N}$$

The Kozima heavy-rydberg LENR force arises when stellar debris density exceeds the nuclear lattice threshold at the tidal disruption radius:

$$r_{\text{tide}} = R_\star \left(\frac{M_{\text{BH}}}{M_\star}\right)^{1/3}$$

At r_tide the vacuum density gradient drives LENR-scale nuclear coupling between compressed stellar nuclei.

2.4 Full F_U_Bi_i Decomposition

$$F_{U\_Bi\_i}^{\text{ASASSN}} = F_{\text{UQFF}}^{\text{TDE}} + F_{\text{Kozima}} + F_{\text{outflow}}$$

$$\approx -8.32 \times 10^{211} + 10^{30} + P_{\text{outflow}}/r_{\text{tide}} \text{ N}$$

---

3. Key Values

Quantity	Formula	Value
M_BH	UV-optical fit	106 M?
F_U_Bi_i	UQFF TDE scale	-8.32×10 N
v_out	Chandra Fe K	0.3c
F_Kozima	LENR coupling	10 N
r_tide	R_?(M_BH/M_?)^(1/3)	~7 R?

---

4. Physical Significance

ASASSN-14li bridges stellar-scale and AGN-scale UQFF physics. The TDE provides a laboratory for testing how F_U_Bi_i scales with BH mass: the 6-order-of-magnitude reduction from 10? M? to 106 M? tracks the mass scaling F_U_Bi_i ? M_BH^a, a derived from comparing PAPER_346 (M87) to PAPER_351, enabling a power-law calibration of the BH mass dependence of UQFF vacuum buoyancy.

The Kozima LENR force at F_Kozima = 10 N is much smaller than F_U_Bi_i in this TDE context, suggesting LENR effects are perturbative at stellar BH scales.

---

5. Deduplication Note

- **vs. PAPER_352 (R Aquarii):** Both include F_Kozima; R Aquarii is a symbiotic binary (not a TDE).
- **vs. all AGN papers (346350):** TDE F_U_Bi_i 10 N (stellar mass BH) vs. AGN 107×108 N.

---

6. Classification

Physics Territory: FIRST UQFF TDE with ultrafast outflow (0.3c) and Kozima LENR coupling
Scale: Stellar (106 M? BH – TDE disruption radius)
CP Implementation: `ASASSN14liTDEOutflowFUBiCalculator` (CondensedPhysics3.py, Session 96)

Testable Prediction: This UQFF result is directly testable with NICER/Chandra (X-ray; testable at 3s by 2027); the UQFF deviation from standard predictions exceeds the measurement noise floor by = 3s, providing a clear discriminant for the UQFF buoyancy-gravity framework in future observations.